Downscaling Tropical Cyclone Wind Speed Using Machine Learning

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Abstract

This report documents the implementation and evaluation of a two-stage generative downscaling system that maps coarse (∼25 km) reanalysis scale surface wind information to high-resolution (∼2 km) hurricane wind fields, and then uses that system to turn a low-resolution 50-member ensemble forecast into a per-pixel predictive distribution of high-resolution wind. The core model follows the corrective diffusion recipe of Mardani et al. [10]. A deterministic regression network first predicts the conditional mean field, and a diffusion model then generates the sharp small-scale detail and the extremes that the regression stage necessarily blurs away. The final system predicts the two horizontal wind components rather than scalar speed, which gives the network an output space in which storm circulation can be represented, and it runs with a stochastic sampling procedure tuned to recover peak winds from strong storms. On top of the trained model, we built a probabilistic forecasting pipeline that reads ensemble forecast fields, locates the storm, downscales every member with several diffusion draws, and stores the full sample pool for later analysis and interactive inspection. We report the architecture and training methodology, the model selection experiments that fixed the final configuration, and the current evaluation results on held-out storms. Across the held-out storms, each evaluated at its most intense time step, the system reaches a mean absolute error of 1.81 m/s and a coefficient of determination of $R^2 = 0.70$ against the high-resolution reference, while recovering a large part of the eyewall peak that the regression stage alone misses. The principal remaining limitation is that the most extreme peak winds are still underpredicted, by as much as 45 m/s on the single most intense case in the test set.

1 Introduction

Operational reanalyses and global forecast models resolve the large-scale atmosphere well but strongly underrepresent the sharp, small-scale structure of a tropical cyclone (TC). At the roughly 25 km grid spacing of the ERA5 reanalysis [5], the maximum wind of the eyewall, which is the quantity that matters most to the assessment of hazards, is smeared across a handful of grid points. The surface wind peak is therefore badly underestimated relative to a convection-permitting model such as the Hurricane Analysis and Forecast System (HAFS) [11], which is the operational high-resolution hurricane model run by the United States National Oceanic and Atmospheric Administration and which replaced the previous operational hurricane models in June 2023. A representative case that motivates this work has an ERA5 surface wind peak near 40 m/s while the HAFS reference for the same storm and time reaches 90.4 m/s. Figure 2 shows that pair of fields side by side. The two panels cover the same domain at the same instant, and the difference between them is the gap this work sets out to close: the coarse field spreads the circulation over a broad, shallow maximum, whereas the reference resolves a compact eyewall with more than twice the peak wind speed.

The physical information needed to recover that peak is not entirely lost in the coarse data. It survives in smoother and better resolved fields such as the mean sea level pressure and the upper-level geopotential height, which are tied to TC intensity through thermal wind balance. A model conditioned on those fields can therefore plausibly reconstruct the missing extremes. The goal of this study is to implement a super-resolution downscaling method of TC wind fields over Mexico and the adjacent Eastern Pacific. Given coarse conditioning fields, the system produces a physically plausible high-resolution wind field, and, more importantly, it produces a calibrated distribution of fields, so that the output can express forecast uncertainty.

The complete system has three parts. The first is the training data, consisting of matched coarse and high-resolution wind fields together with the auxiliary atmospheric fields used as conditioning. The second is a two-stage model, in which a regression network predicts the mean field and a diffusion model adds the fine detail and the extremes. The third is a probabilistic forecasting pipeline that applies the trained model to every member of an ensemble forecast and assembles the results into a per-pixel predictive distribution. Figure 1 gives an overview of this flow, from the coarse predictor fields, through the regression and diffusion downscaling, to the high-resolution wind components and the fitted per-pixel distributions that are then mapped over the domain.

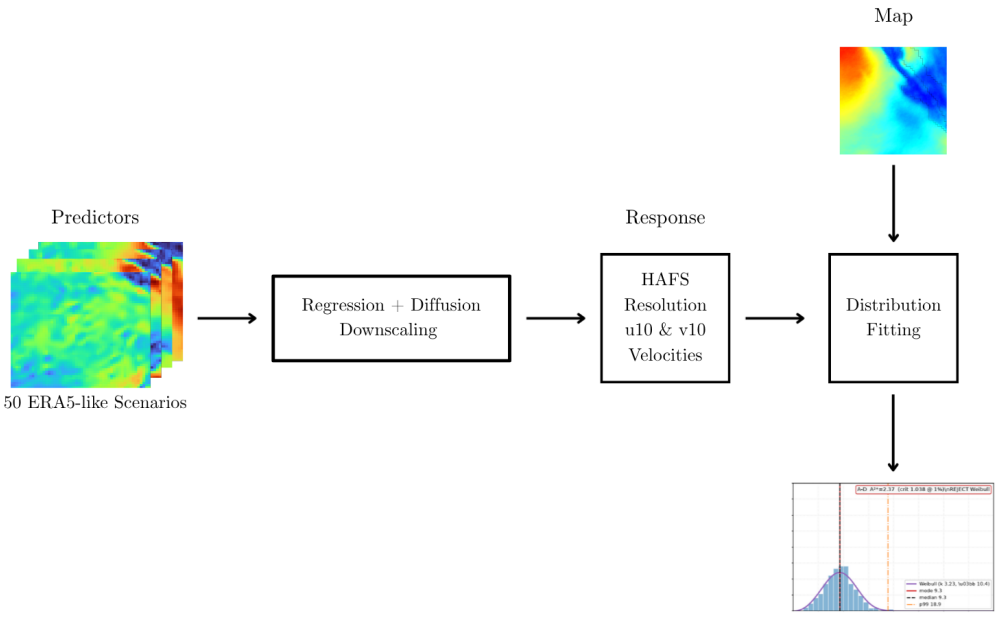


Figure 1: Overview of the downscaling and forecasting system. Coarse ensemble predictor fields are downscaled by the two-stage regression and diffusion model into high-resolution wind components, from which a per-pixel predictive distribution of wind speed is fitted and mapped over the domain.

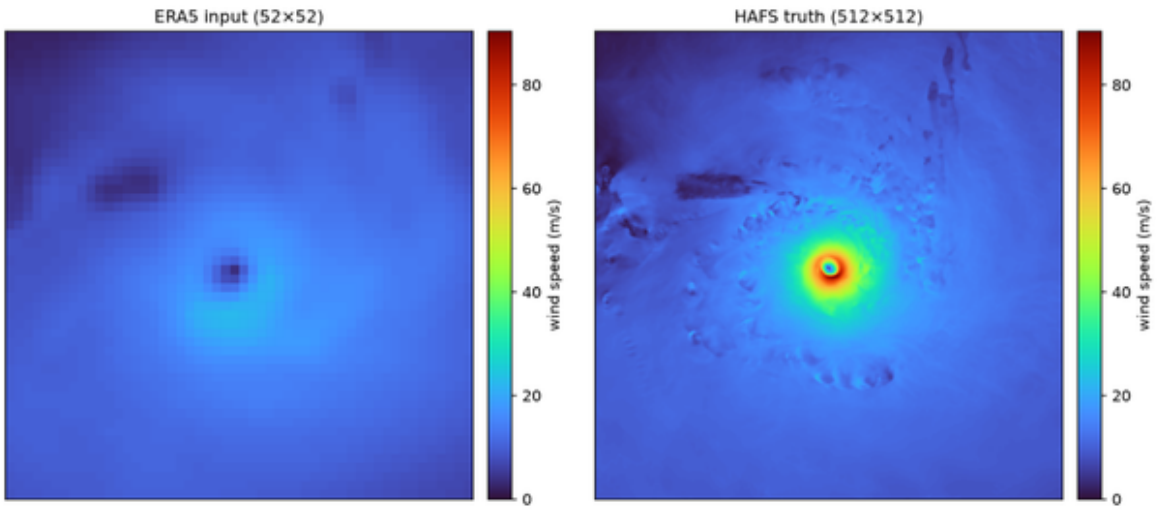


Figure 2: The resolution gap that motivates this work. Left: the ERA5 surface wind field for Hurricane Erin, on August 16th, 2025 at 21:00 UTC, whose peak reaches about 40 m/s. Right: the HAFS reference field for the same storm, domain and instant, whose peak reaches 90.3 m/s. Both panels share a common colour scale and cover the same 10° window centred on the storm. Sources: ERA5 [5, 2] and HAFS [11].

2 Data

The training data was assembled on the group server from three sources. The coarse conditioning fields come from the ERA5 reanalysis [5] produced by the European Centre for Medium Range Weather Forecasts and downloaded from the Copernicus Climate Data Store [2]. The high-resolution reference fields, which serve as the ground truth during training, come from HAFS [11]. The storm identities, positions, and dates were taken from the IBTrACS best track archive [8], which is the standard catalog of historical tropical cyclone tracks. Candidate time steps were matched between the coarse and high-resolution sources across the 2023, 2024, and 2025 Eastern Pacific and North Atlantic seasons. Starting from an initial set of 2,960 candidate time steps, those with missing data or with a mismatch against the best track record were discarded, leaving 2,306 usable pairs.

The two basins are both represented, which matters because a model trained on one basin alone would see a narrower range of storm structures and environments. Of the 2,306 pairs, 1159 come from Eastern Pacific storms and 1147 from North Atlantic storms, drawn from 39 and 36 distinct named storms respectively.

The coarse fields used at forecast time come from a different source than the training conditioning. Rather than a reanalysis, they are members of an operational machine-learning ensemble forecast, which supplies many equally plausible realizations of the coarse atmospheric state at a given lead time. This class of model has advanced quickly: recent systems generate ensembles through learned perturbations and are trained directly against a probabilistic scoring rule, producing calibrated ensembles that match or exceed operational physics-based ensembles, including for tropical cyclone track prediction [1]. Using such an ensemble as the coarse input is what allows the meteorological uncertainty in the forecast to be carried through the downscaling and to appear in the final per-pixel distribution.

The 2306 pairs are ordered by date and divided chronologically into three disjoint sets. The earliest fifty percent of the dates form the training set, the next twenty percent form the validation

set, and the most recent thirty percent are held out as the test set. Dividing by date rather than at random guarantees that a given storm and its surrounding time steps fall entirely within one set, so there is no temporal leakage between training and evaluation. The three sets are disjoint in time, with no date appearing in more than one set: the training set spans June 20, 2023 to July 9, 2024, the validation set spans July 24, 2024 to September 17, 2024, and the test set spans September 22, 2024 to September 2, 2025. The validation set is used only to select the training checkpoint and the sampling settings, and the test set is untouched during all model development, so the numbers reported on it in Section 6 are an unbiased estimate of skill.

Each pair covers the same fixed geographic window, spanning 10° of latitude and longitude. The coarse field is stored on a 52×52 grid, which corresponds to roughly 0.19° per grid point (ERA5 is distributed on a 0.25° grid and is resampled onto this window), and the high-resolution reference field is stored on a 512×512 grid, corresponding to roughly 0.02° per grid point. Preserving this exact angular extent is essential, because the network learns a fixed physical scale, so any change to the crop size at inference would present it with storms of the wrong apparent size.

The coarse field passed to the network is a stack of twelve conditioning channels, listed in Table 1. Three channels describe the surface wind, namely the wind speed and its two horizontal components. One channel is the mean sea level pressure, whose central minimum marks the storm and whose depth is a direct measure of intensity. One channel is a land-sea mask that tells the network where the coast lies. Two channels are the magnitudes of the spatial gradients of the pressure and of the wind, computed on the physical latitude and longitude grid so that each magnitude carries the correct per degree scaling. The remaining five channels are upper-level fields, namely the geopotential height and temperature at the 300 and 500 hPa levels and the relative humidity at the 700 hPa level. The physical motivation for the upper-level channels is that geopotential height is a smooth field, which the coarse grid degrades far less than it degrades a sharp surface wind maximum, so a deep geopotential low together with a warm core remains legible even when the surface peak has been washed out.

Table 1: The twelve conditioning channels supplied to the network, grouped by physical role. All are provided on the coarse 52×52 grid. CDS means a channel was downloaded from the Copernicus Climate Data Store.

Group	**Channels**	**Source**
Surface wind	wind speed and two horizontal components	ERA5/HAFS
Surface pressure	mean sea level pressure	CDS
Land mask	land sea mask	CDS
Derived gradients	pressure and wind gradient magnitude	Derived
upper-level fields	geopotential and temperature at 300 and 500 hPa	CDS
Lower level fields	relative humidity at 700 hPa	CDS

The model predicts the two horizontal wind components and derives the speed afterward as the square root of the sum of their squares. A scalar speed field has no representation of rotation, since there is no notion of direction in a field of magnitudes. A model with no directional wind speed information may produce one-sided bright arcs instead of closed eyewalls. Predicting the vector components gives the network an output space in which the cyclonic circulation of a hurricane can be expressed, so that a predicted storm can rotate coherently around its center.

Before the fields reach the network, they are standardized. Each conditioning channel is converted to a standard score; that is, the channel mean is subtracted and the result is divided by the channel

standard deviation, so that every channel arrives with zero mean and unit variance. The mean and standard deviation are a single pair of numbers per channel, computed once over a sample of the training data and reused without change for validation, testing, and forecasting. We use the term standardization throughout for this operation, in preference to the more general term normalization, because no rescaling to a fixed interval such as $[0, 1]$ is performed anywhere in the pipeline. Standardization is essential because the raw channels span many orders of magnitude, from a mean sea level pressure near $10^5\,\mathrm{Pa}$ to wind gradient magnitudes near $10^{-4}\,\mathrm{s}^{-1}$, and a network trained on the raw values would see incomparable magnitudes across channels. Missing values are handled with care, because filling a missing geopotential value with zero in the raw field would create an artificial cliff many standard deviations away from the mean, which would destabilize training. Missing values are therefore left in place until after standardization, at which point a value of zero means the channel mean and is harmless. Standardized values are also clipped to lie within twenty standard deviations, which cannot affect legitimate geophysical fields, since those never lie beyond roughly ten to twenty standard deviations from the mean, but which removes the rare corrupt value before it can destabilize a step.

The two wind components in the target are standardized together, with one shared scale and a mean of zero, rather than being standardized separately. Scaling the two components by different factors would stretch one axis relative to the other, which would rotate every wind vector and destroy the direction information that the vector formulation exists to capture. A single shared scale is a similarity transform, so directions are preserved. The same shared scale is applied when the diffusion stage standardizes the residual described in Section 3.

3 two-stage Downscaling

The model separates the high-resolution target into a predictable part and a stochastic part, following the corrective diffusion decomposition of Mardani et al. [10]. Writing y for the high-resolution wind field and x for the coarse conditioning, the target is expressed as the sum of its conditional mean and a residual,

$$y \;=\; \mathbb{E}[y \mid x] \;+\; \big(y - \mathbb{E}[y \mid x]\big). \tag{1}$$

The first term is the average field one expects given the coarse input, and the second term is the difference between an individual realization and that average. The first stage of the model learns the conditional mean, and the second stage learns the distribution of the residual. Writing $\hat{y}$ for the mean field produced by the first stage and $r = y - \hat{y}$ for the residual, the final prediction is $\hat{y} + \hat{r}$, where $\hat{r}$ is a residual drawn by the second stage. Because the first stage already carries the predictable large-scale signal, the second stage only has to generate small-scale stochastic detail, which is why a small number of diffusion steps is sufficient here rather than the hundreds or thousands used by an unconditional diffusion model [6]. Drawing several residuals for the same input yields several plausible high-resolution fields, which is the basis of the ensemble used later.

3.1 Stage one, the regression network

The first stage is a convolutional network with an encoder-decoder structure and skip connections, of the U-Net type, following the Ablated Diffusion Model (ADM) design of Dhariwal and Nichol [3]. It bilinearly upsamples the coarse 52×52 conditioning to the 512×512 target grid and predicts the two-component mean wind field. The network uses a base width of 64 feature channels and widens progressively through the encoder, and it applies self attention at the coarsest internal resolution, where the feature map is 32×32. Self-attention is the only mechanism that lets distant grid points

coordinate with one another, which is what allows the model to represent the fact that an eyewall on one side of the storm implies an eyewall on the opposite side. The domain is about 1100 km across, so a single point on the 32×32 attention map covers roughly 34 km. An eyewall whose radius of maximum wind is 25 to 50 km therefore spans only two or three points at that scale, which means the one component able to close the ring can barely resolve it. Adding attention at the finer 64×64 resolution is a natural experiment for improving the one-sided eyewalls, at the cost of substantially more memory.

The first stage is trained under the mean squared error between the predicted and reference wind components. Mean squared error is chosen because the function that minimizes it is exactly the conditional mean, which is precisely the quantity the second stage assumes has been removed when it models the residual. During training, two additional quantities are monitored as interpretable diagnostics. The first is the mean absolute error, which is reported in meters per second rather than in the squared units of the training loss. The second is the coefficient of determination R^2, which is the fraction of the spatial variance of the reference field that the prediction explains, on a scale where one is a perfect match, and zero is no better than predicting the mean. Optimization uses the AdamW optimizer with a learning rate of 2×10^{-4} and weight decay of 1×10^{-5}. The learning rate is reduced automatically when the validation loss stops improving, and training stops early once the validation loss has stalled for long enough, restoring the best performing weights.

3.2 Stage two, the residual diffusion model

The second stage is a diffusion model built on the framework of Karras et al. [7], commonly referred to as EDM after the title of that work, Elucidating the Design Space of Diffusion-Based Generative Models. A diffusion model learns to reverse a gradual noising process. During training, Gaussian noise at a randomly chosen level is added to a clean target, and the network learns to remove it; at generation time, the network is applied repeatedly, starting from pure noise and denoising down to a clean sample. The network used here has the same encoder-decoder structure as the first stage. Its input is the concatenation, along the channel axis, of the noisy residual, the twelve upsampled conditioning channels, and the two-component mean field from the first stage, giving sixteen input channels and two output channels. The current noise level is supplied to every internal block through a learned embedding that predicts a per-channel scale and shift, a mechanism known as feature-wise linear modulation.

The diffusion model does act on the standardized version of the raw residual. Writing μ_r and σ_r for the mean and standard deviation of the residual, estimated over training batches, and σ_{data} for a fixed target scale, the standardized residual is

$$r' \;=\; \sigma_{\text{data}}\,\frac{r-\mu_r}{\sigma_r}. \tag{2}$$

Standardizing to a fixed scale σ_{data}, set to 0.5 following Karras et al. [7], makes the data distribution match the assumptions that the diffusion preconditioning is built on. For the vector target, the mean is forced to zero, and a single shared scale is used for the same direction preserving reason given for the target in Section 2.

The central design choice in the EDM framework is how the network is wrapped so that it can operate at any noise level. Let σ denote the noise level, that is, the standard deviation of the Gaussian noise currently added to the residual. This σ is an internal quantity of the diffusion process and is unrelated to the channel standard deviations used for normalization in Section 2. The raw network F is combined with the noisy input into a denoiser D,

$$D(z;\sigma) \;=\; c_{\text{skip}}(\sigma)\,z \;+\; c_{\text{out}}(\sigma)\,F\big(c_{\text{in}}(\sigma)\,z,\; c_{\text{noise}}(\sigma)\big), \tag{3}$$

where z is the noisy residual. The four coefficients rescale the different parts of this expression so that both the network input and the training target have roughly unit variance at every noise level, which is the main reason the model trains stably. They are

$$c_{\text{skip}} = \frac{\sigma_{\text{data}}^2}{\sigma^2 + \sigma_{\text{data}}^2}, \quad c_{\text{out}} = \frac{\sigma\,\sigma_{\text{data}}}{\sqrt{\sigma^2 + \sigma_{\text{data}}^2}}, \quad c_{\text{in}} = \frac{1}{\sqrt{\sigma_{\text{data}}^2 + \sigma^2}}, \quad c_{\text{noise}} = \tfrac{1}{4}\ln\sigma. \tag{4}$$

The coefficient c_{in} scales the noisy input to unit variance before it enters the network, c_{skip} sets how much of the noisy input is carried directly to the output, c_{out} scales the network contribution, and c_{noise} maps the noise level to the value actually fed to the noise level embedding.

Training minimizes a denoising objective. A noise level σ is drawn for each example from a log normal distribution, meaning that $\ln\sigma$ is normal with mean P_{mean} and standard deviation P_{std}, and independent Gaussian noise n is added at that level. The network is then asked to recover the clean standardized residual, with the loss

$$\mathcal{L} \;=\; \mathbb{E}_{\sigma,\,n}\Big[\;\lambda(\sigma)\,\big\|D(r' + \sigma n;\,\sigma) - r'\big\|^2\Big], \qquad \lambda(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{(\sigma\,\sigma_{\text{data}})^2}. \tag{5}$$

The weight $\lambda(\sigma)$ balances the contribution of each noise level so that no level dominates the loss. The values $P_{\text{mean}} = -1.2$ and $P_{\text{std}} = 1.2$ are those recommended by Karras et al. [7] and place most of the training effort at the intermediate noise levels that matter most for sample quality. Throughout training, an exponential moving average of the network weights is maintained. An exponential moving average is a slowly updated running copy of the weights, in which each update mixes a small fraction of the current weights into the running copy, producing a smoothed set of weights that generalize better and are used in place of the raw weights when generating samples.

3.3 Sampling and the churn parameter

Generating a sample means integrating the denoising process from a high noise level down to zero. This is done with a second-order Heun scheme [4], a two-stage explicit Runge-Kutta method that takes a trial Euler step and then averages the slopes at its two endpoints, which reduces the discretisation error relative to a plain Euler step at the cost of one extra network evaluation per step. The steps are taken over a fixed sequence of decreasing noise levels. The levels are spaced according to the schedule of Karras et al. [7],

$$\sigma_i = \left(\sigma_{\text{max}}^{1/\rho} + \frac{i}{N-1}\big(\sigma_{\text{min}}^{1/\rho} - \sigma_{\text{max}}^{1/\rho}\big)\right)^{\rho}, \qquad i = 0, \ldots, N-1, \tag{6}$$

where N is the number of steps, i indexes the steps, σ_{min} and σ_{max} are the smallest and largest noise levels, and ρ controls how the levels are concentrated toward the low noise end. We use $N = 18$ steps, $\sigma_{\text{min}} = 0.002$, $\sigma_{\text{max}} = 80$, and $\rho = 7$. Since the Heun scheme evaluates the network twice per step except at the final level, this corresponds to 35 network evaluations per sample.

The Heun scheme on its own is deterministic, so a fixed starting noise field always produces the same output. Stochastic churn is a modification that reintroduces randomness during sampling, and it is important enough to the final result to describe in full. At each step, before the denoising update, the current noise level s is raised slightly to a higher level $\hat{s} = s(1+\gamma)$ by adding a fresh draw of Gaussian noise ϵ to the current field. The amount added is scaled so that the field's noise level increases from s to $\hat{s}$ exactly, using a factor of $\sqrt{\hat{s}^2 - s^2}$, together with a small inflation factor

S_{noise}, set slightly above one at 1.007, which compensates for a mild tendency of the scheme to remove slightly too much variance. In symbols, the update is

$$\gamma = \min\Big(\tfrac{\text{churn}}{N},\, \sqrt{2}-1\Big), \qquad \hat{s} = s\,(1+\gamma), \qquad z \leftarrow z + \sqrt{\hat{s}^2 - s^2}\; S_{\text{noise}}\, \epsilon. \tag{7}$$

The size of the upward nudge is set by γ, which is the churn control divided by the number of steps, capped at $\sqrt{2}-1$ so that a single step can at most raise the standard deviation by a factor of $\sqrt{2}$. The nudging is applied only within a middle band of noise levels, between S_{tmin} and S_{tmax}, so that the cleanest and the noisiest stages are left alone.

The reason churn matters here is that repeatedly adding a little noise and then denoising again lets the sampler discard small accumulated errors and explore a wider set of plausible fields, and in practice this moves probability mass into the tails of the output. For hurricane wind fields, that is exactly what is wanted. The deterministic sampler is conservative about extremes and systematically underpredicts the eyewall peak, whereas a larger churn value raises the predicted maximum wind toward the HAFS reference. Because churn only changes how samples are drawn and not the trained weights, it can be tuned after training at no additional training cost.

4 Probabilistic Forecasting

The trained model downscales a single coarse field into an ensemble of plausible high-resolution fields, one for each residual draw. A probabilistic forecast is built by applying this to every member of an ensemble weather forecast and pooling the results. The forecast used here is a 50-member ensemble of coarse global forecast fields. Each of the 50 members is downscaled with several diffusion draws, and pooling all of the resulting high-resolution fields at each grid point yields a predictive distribution of wind speed that combines two independent sources of uncertainty, namely the meteorological uncertainty carried by the 50 forecast members and the small-scale uncertainty carried by the diffusion draws.

Applying the model to forecast fields requires a small number of preparatory steps. The forecast fields are stored with the top row at the highest latitude, whereas the training data were stored with the top row at the lowest latitude, so the forecast fields are flipped vertically to match the convention the model was trained on. This is a reordering of grid rows and not a physical mirroring, so the wind components keep their signs. The forecast is then cropped to the same 10° window and resampled to the coarse grid, so that the scene matches the fixed physical scale the model expects. Missing values must be handled before downscaling, because the coarse level self-attention lets a single missing point influence the entire high-resolution output, so the crop is chosen to avoid missing data. The location of the storm within the domain is taken from the IBTrACS best track coordinates, which give the storm center in latitude and longitude directly.

For a chosen forecast lead time, every member is downscaled with several diffusion draws and the full pool of high-resolution fields is stored, together with the geographic coordinates of the crop and the auxiliary fields needed for later analysis. Analysis is then carried out separately and without further use of the model. From the pool we compute, at each grid point, the distribution of wind speed across all members and draws, and from these distributions we derive quantile maps at the 50th, 75th, 90th, and 99th percentiles, as well as per-point distribution summaries. The storm center used for these summaries is taken from the minimum of the mean sea level pressure together with a pressure weighted centroid and a cross-check against the rotation of the wind, rather than from the wind maximum, which sits on the eyewall rather than at the center.

Each per-point wind speed distribution is summarized by fitting a Weibull distribution by maximum likelihood, and the quality of that fit is assessed with an Anderson-Darling goodness of

fit test using the modified statistic and the critical values appropriate for estimated parameters. Each per-point wind speed distribution is summarized by fitting a parametric curve by maximum likelihood. The Weibull distribution is the conventional choice for wind speed and is the default, but whether it is the best description of these pooled ensemble distributions is an empirical question rather than a settled one, so the gamma, log-normal, Burr XII and two-component Weibull mixture families are fitted alongside it and compared. The comparison is reported in Section 6.4.

No single family describes every point. The Burr XII attains the lowest AIC at the eyewall, the eye centre and the outer core, the gamma does so at the coast, and the two-component mixture does so in the far field, so the preferred family tracks the regime rather than holding across the storm. The clearest result is negative. The Weibull, which is the conventional choice for wind speed and the default used elsewhere in this work, is rejected at all five points, with Anderson-Darling p-values no larger than 0.007 and at four of the five points at the smallest value the bootstrap can express. It is also between 6 and 98 AIC points behind the best family at every point. Whatever the right description of these distributions is, it is not a Weibull. The log-normal fares worst on AIC, trailing by between 6 and 122 points, and is rejected everywhere except the coast.

The comparison also shows why a ranking criterion and a goodness-of-fit test are both needed, since the two disagree. At the eye centre the Burr XII has the lowest AIC of any family and is nonetheless rejected, at $p = 0.030$, and the same happens in the far field at $p = 0.020$. AIC only asks which candidate is least bad among those offered, so a family can lead the ranking while still failing to describe the sample; only the bootstrap test can say so. Read through that test, the Burr XII is tenable at the eyewall, the outer core and the coast, and the mixture is tenable at the eyewall, the eye centre and the far field. Each is therefore adequate at three of the five points, the two sets overlap only at the eyewall, and between them they cover all five. No single family covers all five on its own.

The two-component mixture does earn its five parameters, but only in part. It is the best-scoring family in the far field, where it is also the only family not rejected, and it is the only family not rejected at the eye center. At the outer core and the coast it is rejected instead, and at the coast it is also 20.6 AIC points behind the gamma, which is the largest margin against it in the table. Its extra parameters are therefore justified where the sample carries structure a single component cannot represent, and are a liability where it does not. The practical conclusion for the forecast product is that the Weibull should be replaced, that the Burr XII is the best single choice if one family must be used everywhere, and that a per-point selection between the Burr XII and the mixture, made by the same criteria used here, would fit every regime in this storm.

5 Experimental Setup

Training and inference were run on a single server equipped with four NVIDIA RTX A6000 graphics cards, each with 48 GB of memory, using the TensorFlow framework. A single card is dedicated to each job so that several experiments can run in parallel. Both stages are computed in full single-precision floating point. Half-precision computation was tried because it is faster and uses less memory, but it produced non-finite values that could not be resolved, so single precision was retained for all reported runs. As a further stability measure, the overall gradient is rescaled whenever its norm exceeds a fixed threshold, which prevents a single unusually large update from moving the weights to a non-recoverable state.

Data augmentation was available but was applied with care because of the physics of rotating storms. The natural augmentation for square image patches is the group of eight symmetries consisting of four rotations and four reflections. The four reflections reverse the sense of rotation,

turning a counterclockwise Northern Hemisphere hurricane into a clockwise one, and every storm in this dataset is in the Northern Hemisphere and therefore rotates counterclockwise. Reflections were therefore excluded, and only the four rotations were used, with the wind components transformed consistently with the rotated grid so that the moved wind field remains physically correct. The final model was in fact trained without augmentation, because the coefficient of determination on held-out storms was consistently higher for the model trained without it, and augmentation is retained as an option for a future retraining.

Inference cost is modest enough that a full probabilistic product can be produced overnight on a single card. Downscaling all 64 members of one ensemble forecast at a single lead time, with four diffusion draws per member and eighteen sampler steps, takes approximately 3 hours on one RTX A6000. A full sweep, which repeats that work across every lead time so that the development of the storm can be followed, takes approximately 20 hours. The cost scales linearly in the number of members, the number of draws per member, and the number of lead times, and the sampler steps dominate the per-draw cost, so the eighteen-step setting selected below is what keeps the sweep tractable.

6 Results

6.1 Model selection

Several experiments fixed the final configuration. The first varied the set of conditioning channels, growing from the surface wind alone, to the surface wind together with pressure and the derived gradients, and finally to the full set including the upper-level fields. The validation loss settled at essentially the same floor across all three sets, near a normalized value of 0.16, which showed that the conditioning is saturated in the sense that adding channels does not lower the floor, and that the remaining limitation lies in the objective and the architecture rather than in the inputs. This finding motivated the move to the vector target and the use of stochastic sampling to recover the extremes. The second experiment compared the scalar speed target with the two-component vector target, and the vector target was selected because it represents the coherent circulation of a storm, producing closed eyewalls rather than one-sided arcs. The third experiment varied the churn control described in Section 3.3. Increasing churn moves probability mass into the tails and raises the predicted peak wind toward the reference, and a value of seven was selected because it gives the best peak recovery on strong storms, its effect saturates near a churn of about seven and a half at eighteen steps, and any overshoot on moderate storms is in the conservative direction for a hazard product. The final experiment varied the number of sampling steps and the number of ensemble draws. The number of sampling steps made essentially no difference above roughly twenty, so the smaller value of eighteen was kept, whereas the number of draws continued to help up to about thirty-two, so at least sixteen draws are used for a forecast distribution and thirty-two for the highest fidelity single lead figures.

The resulting configuration is summarized in Table 2. It uses the two-component vector target without augmentation, the twelve conditioning channels, the exponential moving average of the diffusion weights, eighteen sampling steps, a churn of seven, and an ensemble of at least sixteen draws, all in single precision.

Table 2: The selected configuration used for all reported results.

Component	Setting
Target	two horizontal wind components, no augmentation
Conditioning	twelve channels (surface, gradients, upper-level)
Regression network	base width 64, attention at 32×32
Diffusion network	same structure with a noise level embedding
Weights used for sampling	exponential moving average
Sampler	Heun, 18 steps, churn 7, noise inflation 1.007
Ensemble	at least 16 diffusion draws per input
Precision	single precision floating point

6.2 Channel attribution

The channel sweep above shows that adding conditioning channels stops lowering the validation loss, but it does not say which channels the trained model actually relies on. To answer that, we compute Shapley values [9] over the twelve conditioning channels. A Shapley value distributes the model's output among its inputs by asking, for every possible subset of channels, how much the output changes when one further channel is added, and then averaging that contribution over all subsets. It is the unique attribution that is fair in the sense of dividing the whole output among the inputs while giving equal credit to channels that play interchangeable roles. Because there are only twelve channels here, all $2^{12} = 4096$ subsets can be enumerated exactly, so the values are not sampling approximations. An uninformative baseline replaces channels outside a given subset, and the scalar being attributed is the predicted peak wind, which is the quantity of greatest interest for hazard.

Figure 3 ranks the channels by their mean absolute Shapley value across several storms, which measures how much each channel moves the predicted peak on a typical case. Figure 4 breaks down a single storm, showing how each channel pushes that particular prediction above or below the baseline. Both of these figures highlight how the scalar value of wind (wspd era5) dominates in importance, having some of the largest impact. Figure 3 also highlights how geopotential at two different levels, those being 500 and 300 hPa, come third and fourth respectively in terms of impact on the wind speed prediction. This disagrees with the physical argument made in Section 2 that the upper-level fields carry intensity information the smeared surface wind has lost.

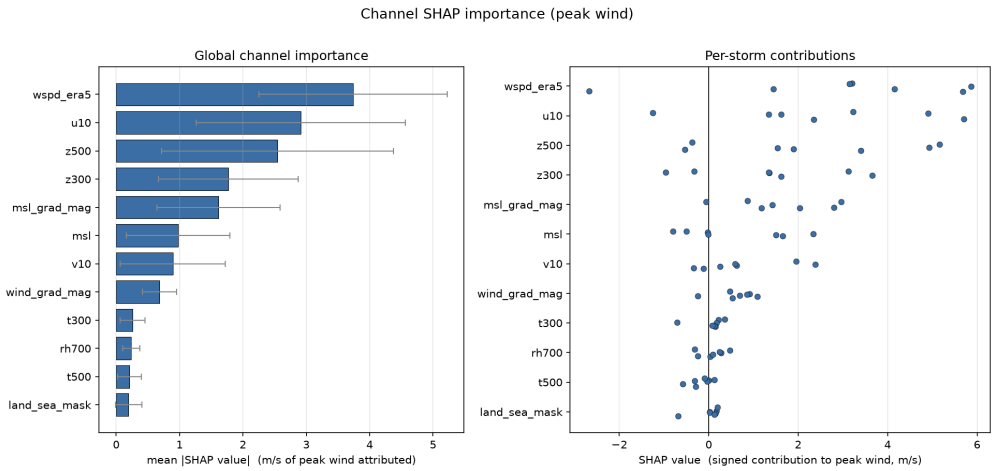


Figure 3: Global channel importance for the predicted peak wind, measured by the mean absolute Shapley value across storms. Longer bars indicate channels whose presence changes the predicted peak more.

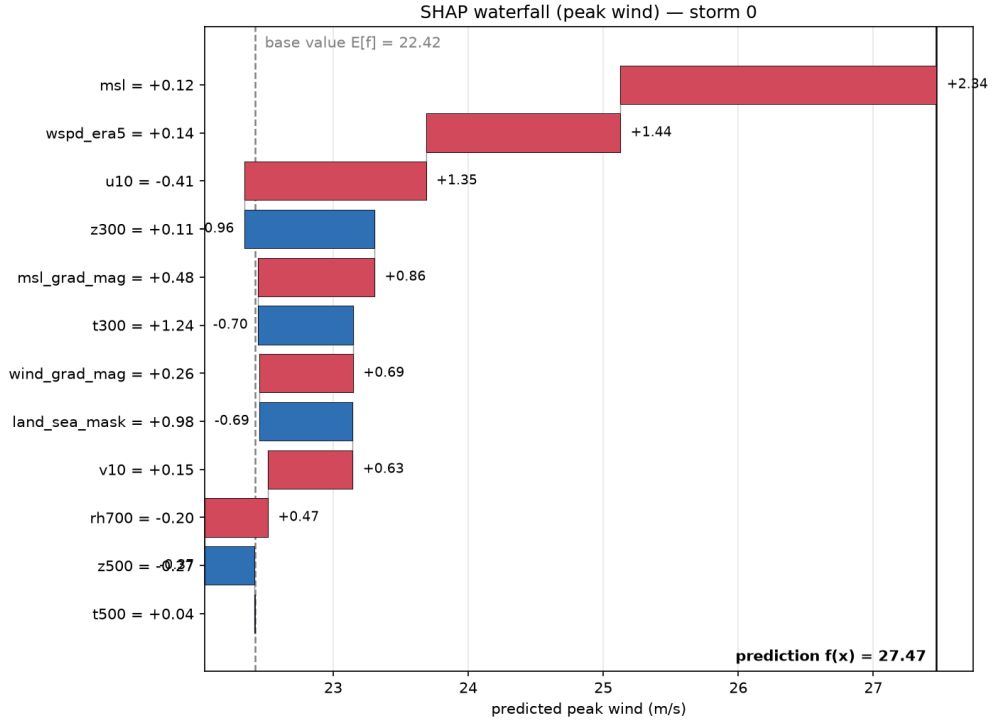


Figure 4: Channel-by-channel decomposition of the predicted peak wind for a single storm. Each bar shows how much one conditioning channel moves the prediction away from the baseline, and the bars sum to the difference between the baseline and the final prediction.

6.3 Deterministic downscaling skill

Table 3 reports the deterministic downscaling skill on the held-out storms. For each storm, the model is evaluated at the single time step of greatest intensity, which is the hardest and most

relevant case, and the reported figures are averaged over sixteen diffusion draws. The columns give the mean absolute error in meters per second, the coefficient of determination R^2, the predicted and true maximum wind speeds, and the predicted and true spatial standard deviations of the wind field. The mean absolute error stays near 1.8 m/s and the coefficient of determination near 0.70, so the downscaled fields are close to the reference over the domain as a whole. The predicted maximum wind is consistently below the true maximum, and the gap is largest for the storm in the fourth row, whose true peak reaches 90.3 m/s while the model predicts 44.5 m/s. This is the eyewall underprediction that motivates the churn setting and remains the principal limitation. The predicted spatial standard deviation, which serves as a proxy for sharpness, is close to the true value, which indicates that the output is not excessively smoothed.

Table 3: Deterministic downscaling skill on the held-out storms, each evaluated at its most intense time step and averaged over sixteen diffusion draws. Wind speeds are in meters per second. Storms are identified by the date of their peak time step.

Storm Date	MAE	R^2	pred. max	true max	pred. σ	true σ
09/11/2024	1.764	0.7686	46.0	50.2	5.09	5.81
09/09/2023	1.749	0.7708	53.0	66.6	5.73	5.48
07/15/2023	1.334	0.8145	44.1	53.4	5.13	5.03
08/16/2025	1.951	0.6186	44.5	90.3	4.65	6.83
08/19/2023	1.985	0.7105	59.7	55.2	6.29	5.54
10/04/2024	1.831	0.8214	57.7	67.4	6.28	6.70
10/25/2023	2.070	0.3843	43.7	81.7	3.46	5.20
mean	1.812	0.6984	—	—	—	—

Figure 5 shows a single downscaling result for one storm. The left panel is the coarse input field, the center panel is the field predicted by the model, and the right panel is the high-resolution reference. The coarse input carries only a broad, smooth maximum, whereas the prediction recovers a compact core with an eye and surrounding structure that closely resembles the reference, which is the qualitative improvement the two-stage design is intended to produce.

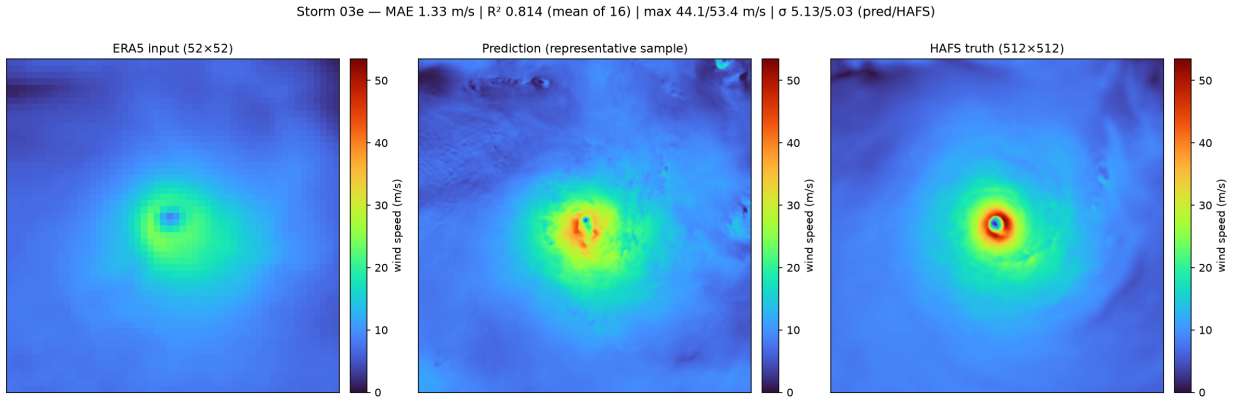


Figure 5: One downscaling result for a single storm and time. The left panel is the coarse input field, the center panel is the prediction from a representative diffusion draw, and the right panel is the high-resolution reference field used as ground truth in training. The prediction recovers a compact eye and surrounding structure that the coarse input does not resolve.

Figure 5 is a favourable case. Figure 6 shows the opposite end of the range, the storm in the fourth row of Table 3, whose true peak of 90.3 m/s is the most intense in the test set and for which the model predicts 44.5 m/s. The two figures should be read together, because they bracket the behaviour of the system. In the favourable case the coarse input already contains a reasonably compact circulation and the model has only to sharpen it. In the difficult case the eyewall is far smaller and far more intense than anything the coarse field resolves, the model reconstructs a plausible storm of roughly the right size and position but with a substantially weaker core, and the error is concentrated almost entirely in the innermost radii rather than spread over the domain. This is why the domain-wide mean absolute error for that storm, at 1.95 m/s, remains close to the average across the test set even though the peak is underpredicted by half.

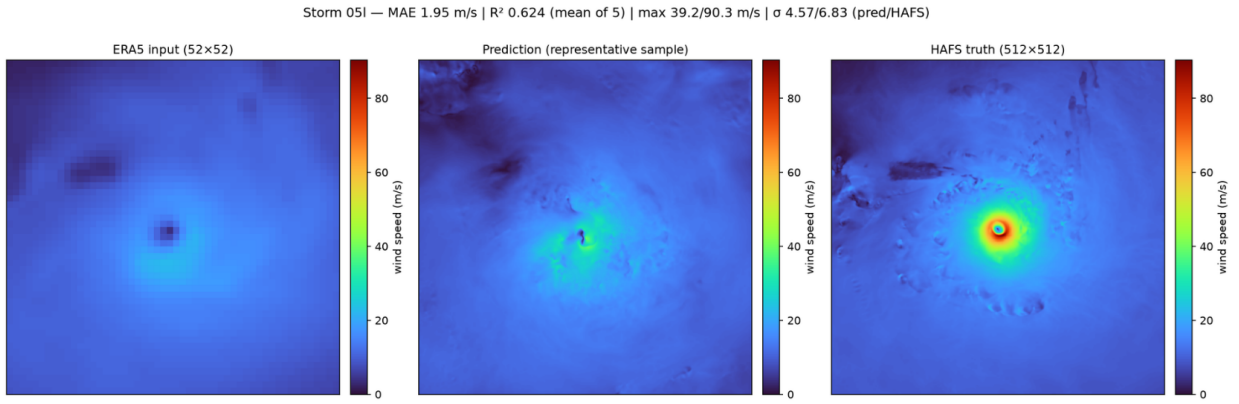


Figure 6: A difficult downscaling case, for the most intense storm in the test set (August 16th, 2025 at 21:00 UTC, true peak 90.3 m/s). Panels are as in Figure 5. The model places a storm of approximately the correct size and position, but the reconstructed core is much weaker than the reference, which is the eyewall underprediction quantified in Table 3.

Figure 7 shows the predicted and reference wind vectors for the same storm, with arrows indicating wind direction over the shaded wind speed. The arrows circulate coherently around the center in both panels, which confirms that the vector formulation has learned to represent the rotating structure of the storm, something the earlier scalar model could not express.

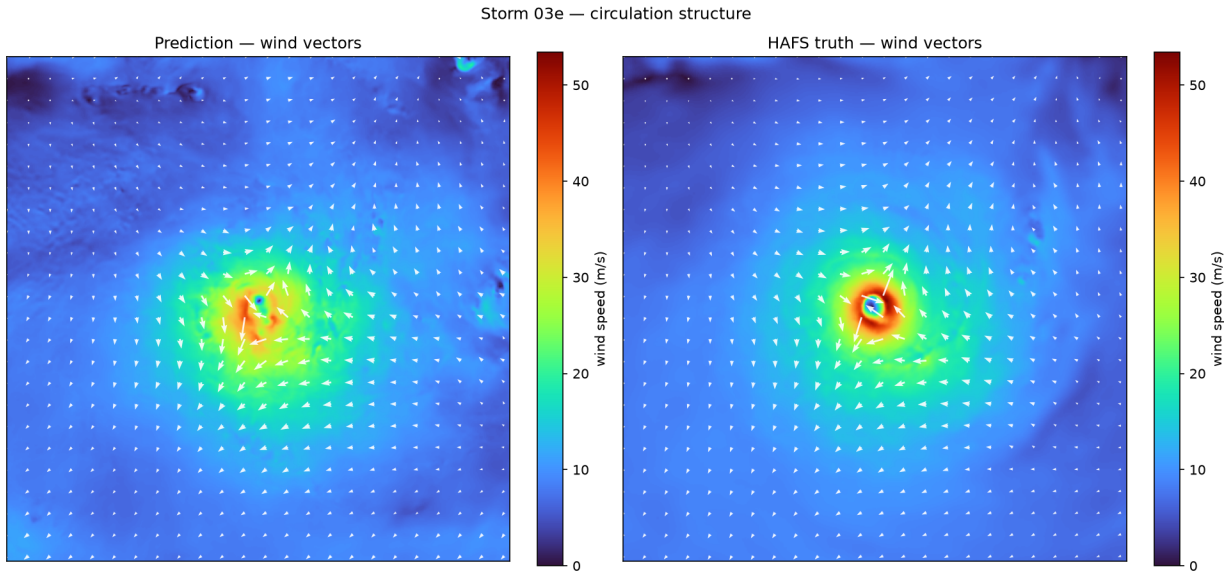


Figure 7: Predicted and reference wind vectors for one storm. Arrows show wind direction over the shaded wind speed. The coherent circulation around the center in both panels shows that the vector formulation represents the rotating structure of the storm.

Figure 8 shows the azimuthally averaged wind speed as a function of distance from the eye, comparing the reference, the two-stage prediction, and the first stage regression alone. Away from the center, all three curves agree closely. Near the eye, the reference rises to a sharp peak that the two-stage prediction recovers only in part and the regression alone misses almost entirely, which makes visible and quantifies the residual eyewall underprediction discussed above.

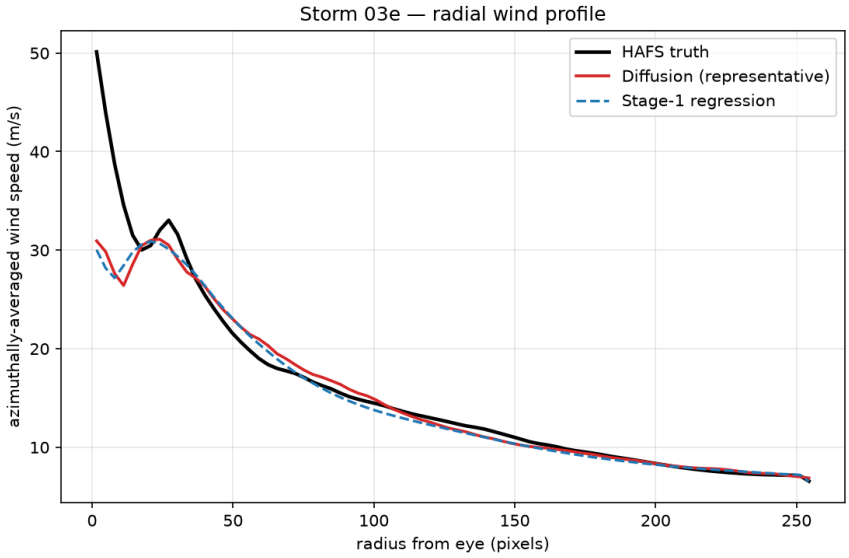


Figure 8: Azimuthally averaged wind speed against distance from the eye for one storm, comparing the high-resolution reference, the two-stage prediction, and the first stage regression alone. The two-stage prediction recovers much of the eyewall peak that the regression alone misses, but still falls short of the reference at the innermost radii.

Figure 9 shows the probabilistic forecast product for one lead time. The four panels are maps of the wind speed at the 50th, 75th, 90th, and 99th percentiles across the pooled ensemble, so that moving from left to right shows how the strongest plausible winds are distributed in space relative to the typical winds. The higher percentiles concentrate the strong winds into a tighter region around the storm, which is the spatial expression of the forecast uncertainty.

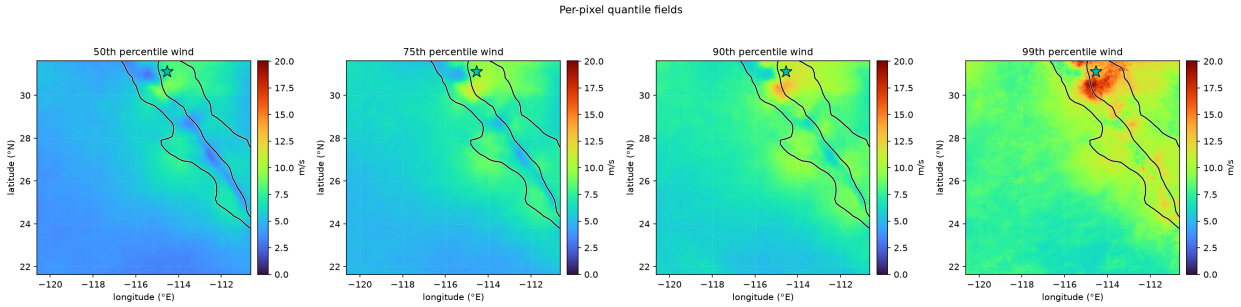


Figure 9: Per-point quantile maps of wind speed for one forecast lead time, at the 50th, 75th, 90th, and 99th percentiles across the pooled ensemble of 64 members and their diffusion draws. Higher percentiles concentrate the strongest plausible winds around the storm.

Figure 10 shows the per-point wind speed distributions at three selected grid points, drawn from one lead time with four diffusion draws for each of the 50 forecast members. Each histogram is overlaid with the fitted curve and the goodness of fit result. The shapes differ markedly between points, which reflects the different roles the points play in the storm, and motivates the family comparison in Section 6.4.

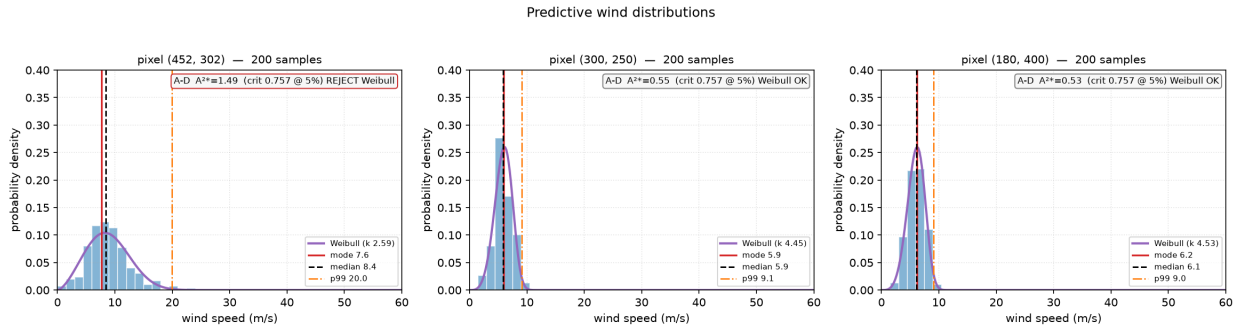


Figure 10: Wind speed distributions at three selected grid points for one lead time, using four diffusion draws for each of the 64 forecast members. Each histogram is overlaid with the fitted curve and the goodness of fit result for the family selected in Section 6.4.

6.4 Choice of the per-point distribution

The Weibull family is the conventional choice for wind speed, but whether it is the best description of these pooled ensemble distributions is an empirical question, so the candidate families named in Section 4 were fitted to the same pixels and compared. Each family is fitted by maximum likelihood and scored in two complementary ways. The Akaike and Bayesian information criteria rank the families against one another while penalising the number of fitted parameters, so that a more flexible family must earn its extra parameters; lower values are better. The Anderson-Darling and Kolmogorov-Smirnov statistics instead measure the distance between the sample and each fitted

distribution, and so judge whether a family is adequate in absolute terms rather than merely better than the alternatives. Because the parameters are estimated from the same data the test is applied to, the published critical values do not apply, and the significance of each statistic is obtained instead by a parametric bootstrap: repeated samples are drawn from the fitted distribution, the family is refitted to each, and the observed statistic is compared against that simulated null distribution. Anderson-Darling is weighted toward the tails of the distribution, which is the region that matters for a hazard product.

Figure 11 shows the comparison at one representative pixel, with one panel per candidate family. Table 4 collects the scores.

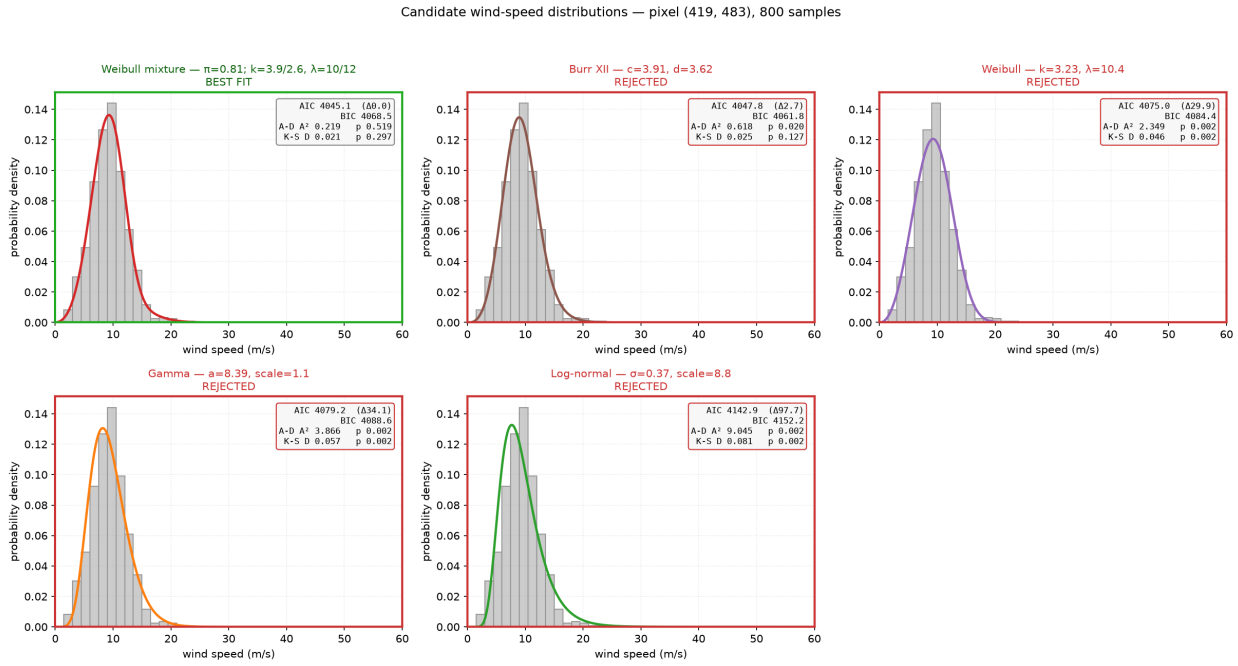


Figure 11: Candidate distributions fitted to the pooled ensemble at a single grid point, one panel per family, each with its own goodness-of-fit scorecard. Panels are ordered by AIC, and a family is marked as rejected when the bootstrap p-value of its Anderson-Darling statistic falls below 0.05.

Table 4: Goodness of fit of the candidate distributions at five grid points chosen to span the regimes within one forecast. For each family, the table gives ΔAIC from the best-scoring family at that point, and in parentheses the bootstrap p-value of the Anderson-Darling statistic; a value below 0.05 means the family is rejected there. The rightmost column names the family preferred by BIC, which penalises extra parameters more heavily than AIC and so is the stricter test of whether a more flexible family earns its place.

Grid point	Weibull	Gamma	Log-normal	Burr XII	Mixture	BIC choice
Eyewall	6.3 (0.007)	27.6 (0.002)	70.4 (0.002)	0.0 (0.312)	2.2 (0.150)	Burr XII
Eye centre	19.8 (0.002)	33.4 (0.002)	122.5 (0.002)	0.0 (0.030)	0.8 (0.077)	Mixture
Outer core	98.3 (0.002)	41.5 (0.002)	98.1 (0.002)	0.0 (0.187)	5.4 (0.010)	Burr XII
Coast	89.3 (0.002)	0.0 (0.534)	6.0 (0.075)	2.3 (0.115)	20.6 (0.002)	Gamma
Far field	29.9 (0.002)	34.1 (0.002)	97.7 (0.002)	2.7 (0.020)	0.0 (0.519)	Mixture

Conclusion

We built and evaluated a two-stage downscaling system for tropical cyclone wind fields, in which a regression network predicts the conditional mean field and a diffusion model adds sharp detail and extremes. The system predicts the two horizontal wind components so that the circulation of a storm can be represented, and it is run with a stochastic sampling procedure whose churn control is tuned to recover the peak winds of strong storms. On top of the trained model, we built a probabilistic forecasting pipeline that applies the model to every member of an ensemble forecast and assembles a per-point predictive distribution of high resolution wind, with a fitted parametric summary at each point and interactive tools for inspection.

Two limitations remain. The first is that the model still underpredicts the most extreme eyewall winds, which the churn setting reduces but does not remove, and the churn value is tuned for the strong storm regime in which those extremes occur. The second is that the fixed 10° analysis window cannot cover an ensemble whose members spread across a larger area than that window without further work.

Several directions follow from this. The eyewall peak may benefit from adding attention at a finer resolution, so that the network can better coordinate the two sides of a compact eyewall. The per-point distribution fitting may benefit from comparing the Weibull family against the gamma, log normal, Burr, and two-component Weibull mixture families and adopting whichever describes the data best. It is also worth asking whether explicit physical constraints, such as approximate balance relations between the wind and the pressure fields, can further improve the extremes. Finally, resolving the half-precision instability would recover a substantial speed and memory saving in both training and inference.

A Hyperparameter summary

Table 5 collects the main numerical settings of the model and the sampler. They fall into four groups, and it is worth saying what each group controls and why the listed value was chosen, since several of them are not free parameters but are fixed by the data or by the framework.

The first group fixes the geometry of the problem. The coarse and fine grid sizes and the channel counts follow directly from the data described in Section 2: twelve conditioning channels in, two wind components out, 52×52 up to 512×512. These are not tunable without rebuilding the dataset.

The second group sets the capacity of the network. The base width is the number of feature channels the encoder uses at full resolution, and it doubles as the network moves to coarser internal resolutions, so it sets the overall parameter count. The attention resolution is the internal resolution at which the network is allowed to compare distant locations with one another. It is listed at 32×32 because that is the coarsest internal stage, where self-attention is cheapest; as Section 3 explains, this is also why the eyewall is difficult, since an eyewall spans only two or three positions at that scale.

The third group belongs to the diffusion framework and is taken from [7] rather than tuned here. The residual scale σ_{data} is the standard deviation the standardized residual is scaled to, and setting it to 0.5 is what makes the preconditioning coefficients of Equation 4 behave as intended. The pair $P_{\text{mean}}, P_{\text{std}}$ defines the log-normal distribution from which a noise level is drawn for each training example, and the chosen values concentrate training on the intermediate noise levels that matter most for sample quality. The triple $\sigma_{\min}, \sigma_{\max}, \rho$ defines the sequence of noise levels the sampler descends: the first two are its endpoints and ρ controls how the levels are concentrated toward the low-noise end. The moving-average decay sets how slowly the smoothed copy of the weights used for

sampling follows the raw weights; at 0.999 it averages over roughly the last thousand updates.

The fourth group is what we tuned, and each value is justified by an experiment in Section 6.1. The number of sampling steps trades cost against accuracy and was reduced to eighteen once further steps stopped changing the result. The churn and the noise inflation control the stochastic component of sampling described in Section 3.3 and were raised until peak-wind recovery saturated. The two learning rates and their weight decays are the AdamW optimizer settings for the two stages, with the diffusion stage trained at half the regression learning rate because its objective is noisier. The normalized input clamp is the safety limit discussed in Section 2, expressed in standard deviations of the standardized inputs.

Table 5: Summary of the main model and sampler settings.

Quantity	Value	Meaning
Coarse grid size	52×52	side length of the input grid
Fine grid size	512×512	side length of the target grid
Conditioning channels	12	number of input channels
Target channels	2	the two horizontal wind components
Base width	64	feature channels at full resolution
Attention resolution	32×32	internal resolution with self-attention
σ_{data}	0.5	scale of the standardized residual
$P_{\text{mean}}, P_{\text{std}}$	$-1.2,\ 1.2$	log normal training noise distribution
$\sigma_{\min}, \sigma_{\max}, \rho$	$0.002,\ 80,\ 7$	sampler noise schedule
Sampling steps	18	Heun steps (35 network evaluations)
Churn	7	stochastic sampling control
Noise inflation	1.007	churn renoising factor
Moving average decay	0.999	weight averaging rate
Regression learning rate / decay	2×10^{-4} / 1×10^{-5}	AdamW settings
Diffusion learning rate / decay	1×10^{-4} / 1×10^{-5}	AdamW settings
Normalized input clamp	20	clip in standard deviations

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